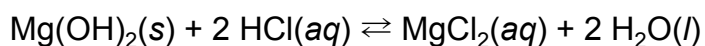


Gastroesophageal reflux disease (GERD) is a condition that occurs when acidic gastric juice flows upward from the stomach into the esophagus. This disease is a result of the lower esophageal sphincter's (LES) inability to close. When functioning properly, the LES, which separates the stomach from the esophagus, opens when food is ingested and then closes. If the LES fails to close, gastric juice rushes back into the esophagus, causing tissue irritation and damage.

Antacids are medications that relieve symptoms of GERD by reducing the acidity of the gastric juice, which on average contains 5×10^{-4} moles of $\text{HCl}(aq)$ per 50 mL of gastric juice. A typical antacid contains the water insoluble compounds $\text{Mg}(\text{OH})_2(s)$ and $\text{CaCO}_3(s)$. Both compounds neutralize the acid present in the gastric juice, as shown in Reactions 1 and 2.



Reaction 1



Reaction 2

The neutralizing capacity of an antacid can be determined by dissolving an antacid tablet in an excess amount of $\text{HCl}(aq)$ of known concentration and titrating the unreacted $\text{HCl}(aq)$ with $\text{NaOH}(aq)$. In an experiment, an antacid tablet was placed in a flask containing 50 mL of 1.5 M $\text{HCl}(aq)$. After the antacid tablet had completely reacted, the solution was gently heated to remove any $\text{CO}_2(g)$. The degassed solution was then titrated using 0.50 M $\text{NaOH}(aq)$ in the presence of an indicator. Figure 1 shows the results obtained from the titration.

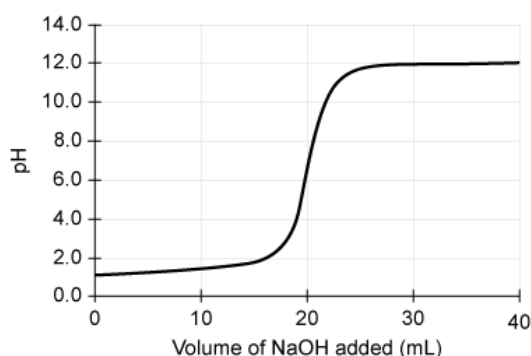


Figure 1 Titration of excess $\text{HCl}(aq)$ with $\text{NaOH}(aq)$

The total number of moles of $\text{HCl}(aq)$ initially present in the beaker, $n_{\text{HCl total}}$, is equal to the sum of the number of moles of $\text{HCl}(aq)$ neutralized by the antacid tablet, $n_{\text{HCl,antacid}}$ and the number of moles of excess $\text{HCl}(aq)$ neutralized during the titration by $\text{NaOH}(aq)$, $n_{\text{HCl,titration}}$.

$$n_{\text{HCl total}} = n_{\text{HCl,antacid}} + n_{\text{HCl,titration}}$$

Equation 1

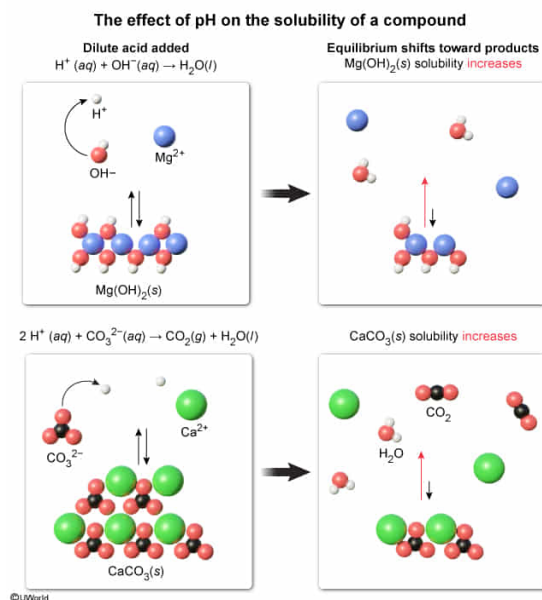
Dissolving the antacid sample in HCl(aq) :

- ☐ A. increases the solubility of $\text{CaCO}_3(\text{s})$ and decreases the solubility of $\text{Mg(OH)}_2(\text{s})$. [5%]
- ☐ B. decreases the solubility of $\text{CaCO}_3(\text{s})$ and increases the solubility $\text{Mg(OH)}_2(\text{s})$. [6%]
- ✓ ☒ C. increases the solubility of both $\text{CaCO}_3(\text{s})$ and $\text{Mg(OH)}_2(\text{s})$. [72%]
- ☐ D. has no effect on the solubility of $\text{CaCO}_3(\text{s})$ and $\text{Mg(OH)}_2(\text{s})$. [15%]

Correct

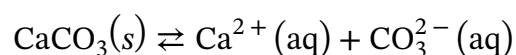
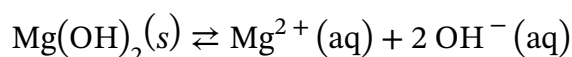
72% answered correctly

Explanation:



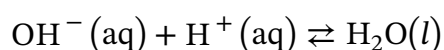
Le Châtelier's principle says that if the **equilibrium state** of a system is disturbed, the system will shift toward the reactants or the products to counteract the disturbance and establish a new equilibrium. **Removing products** causes the reaction to **form more products** (ie, the equilibrium shifts to the right) whereas **adding products** causes the reaction to **form more reactants** (ie, the equilibrium shifts to the left).

When an antacid tablet is dissolved in an aqueous solution, the following dissolution reactions occur:

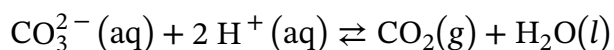


Because both $\text{OH}^{-}(\text{aq})$ and $\text{CO}_3^{2-}(\text{aq})$ are bases, they **react** with acids. Consequently, when the antacid tablet is dissolved in an aqueous solution of HCl (a source of H^{+}), some of the OH^{-}

(aq) reacts with $\text{H}^{+}(\text{aq})$ to form $\text{H}_2\text{O}(\text{l})$:



Some of the $\text{CO}_3^{2-}(\text{aq})$ also reacts with $\text{H}^{+}(\text{aq})$ to form $\text{H}_2\text{O}(\text{l})$ and $\text{CO}_2(\text{g})$:



The decrease in the concentrations of both $\text{OH}^-(\text{aq})$ and $\text{CO}_3^{2-}(\text{aq})$ causes the dissolution equilibria to shift towards the products to form more $\text{OH}^-(\text{aq})$ and $\text{CO}_3^{2-}(\text{aq})$. Therefore, both $\text{Mg}(\text{OH})_2(\text{s})$ and $\text{CaCO}_3(\text{s})$ dissolve more readily (ie, solubility increases) in the presence of $\text{HCl}(\text{aq})$.

(Choices A, B, and D) $\text{HCl}(\text{aq})$ is an acid and reacts with $\text{OH}^-(\text{aq})$ and $\text{CO}_3^{2-}(\text{aq})$, causing the solubility of both $\text{Mg}(\text{OH})_2(\text{s})$ and $\text{CaCO}_3(\text{s})$ to increase as the dissolution equilibrium shifts toward products.

Educational objective:

Le Châtelier's principle states that if a system at equilibrium is disturbed, the system will shift toward the reactants or the products to counteract the disturbance, and a new equilibrium will be established. The solubility of base-containing compounds increases in the presence of an acid by shifting the dissolution equilibrium toward products.

Time spent: 0 sec

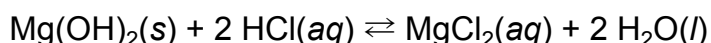
Subject:
General Chemistry

QID: 403932

System:
Solutions and Electrochemistry

Gastroesophageal reflux disease (GERD) is a condition that occurs when acidic gastric juice flows upward from the stomach into the esophagus. This disease is a result of the lower esophageal sphincter's (LES) inability to close. When functioning properly, the LES, which separates the stomach from the esophagus, opens when food is ingested and then closes. If the LES fails to close, gastric juice rushes back into the esophagus, causing tissue irritation and damage.

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Reaction 1



Reaction 2

The neutralizing capacity of an antacid can be determined by dissolving an antacid tablet in an excess amount of $\text{HCl}(aq)$ of known concentration and titrating the unreacted $\text{HCl}(aq)$ with $\text{NaOH}(aq)$. In an experiment, an antacid tablet was placed in a flask containing 50 mL of 1.5 M $\text{HCl}(aq)$. After the antacid tablet had completely reacted, the solution was gently heated to remove any $\text{CO}_2(g)$. The degassed solution was then titrated using 0.50 M $\text{NaOH}(aq)$ in the presence of an indicator. Figure 1 shows the results obtained from the titration.

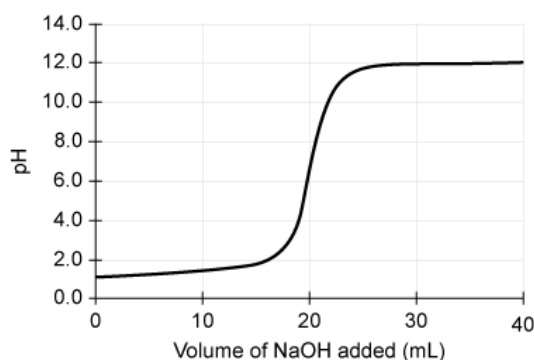


Figure 1 Titration of excess $\text{HCl}(aq)$ with $\text{NaOH}(aq)$

The total number of moles of $\text{HCl}(aq)$ initially present in the beaker, $n_{\text{HCl, total}}$, is equal to the sum of the number of moles of $\text{HCl}(aq)$ neutralized by the antacid tablet, $n_{\text{HCl, antacid}}$ and the number of moles of excess $\text{HCl}(aq)$ neutralized during the titration by $\text{NaOH}(aq)$, $n_{\text{HCl, titration}}$.

$$n_{\text{HCl, total}} = n_{\text{HCl, antacid}} + n_{\text{HCl, titration}}$$

Equation 1

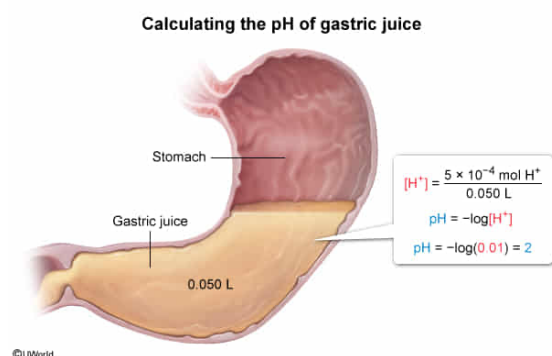
What is the average pH of gastric juice as described in the passage?

- ✓ ☒ A. 2 [67%]
☐ B. 3 [20%]
☐ C. 5 [8%]
☐ D. 7 [2%]

Correct

67% answered correctly

Explanation:



The **pH scale** indicates the acidity of a solution, and a **pH value** is directly related to the **molar concentration** (ie, molarity, M) of $H^+(aq)$ in the solution as defined by:

$$pH = -\log[H^+]$$

The passage states that on average 50 mL of gastric juice contains 5×10^{-4} moles of $H^+(aq)$. Dividing the moles of $H^+(aq)$ by the volume of gastric juice (in liters) yields the molar concentration of $H^+(aq)$:

$$\frac{5 \times 10^{-4} \text{ moles } H^+}{50 \text{ mL}} \times \frac{1000 \text{ mL}}{1 \text{ L}} = 0.01 \text{ M } H^+$$

Therefore, the average pH of the gastric juice described in the passage is:

$$pH = -\log(0.01) = 2$$

Without a calculator, the negative logarithm can be approximated using the formula:

$$-\log(m \times 10^{-n}) \approx n - 0.m$$

where n is a whole number and m is a number between 1 and 10. As such, the average pH of the gastric juice can be approximated as:

$$pH = -\log(1 \times 10^{-2}) \approx 2 - 0.1$$
$$pH \approx 1.9 \approx 2$$

(Choice B) A pH of 3 is obtained if the *moles* of $H^+(aq)$ are used in the calculation rather than the *molar concentration* (mol/L).

(Choice C) If the units of milliliters are not converted to liters, a pH of 5 is obtained.

(Choice D) A pH of 7 is neutral. For an acidic solution, $\text{pH} < 7$.

Educational objective:

The pH of a solution is defined as $\text{pH} = -\log[\text{H}^+]$, where $[\text{H}^+]$ is the H^+ concentration in units of molarity (moles/liter).

Time spent:0 sec

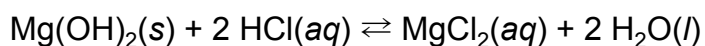
Subject:
General Chemistry

QID:403933

System:
Solutions and Electrochemistry

Gastroesophageal reflux disease (GERD) is a condition that occurs when acidic gastric juice flows upward from the stomach into the esophagus. This disease is a result of the lower esophageal sphincter's (LES) inability to close. When functioning properly, the LES, which separates the stomach from the esophagus, opens when food is ingested and then closes. If the LES fails to close, gastric juice rushes back into the esophagus, causing tissue irritation and damage.

Antacids are medications that relieve symptoms of GERD by reducing the acidity of the gastric juice, which on average contains 5×10^{-4} moles of $\text{HCl}(aq)$ per 50 mL of gastric juice. A typical antacid contains the water insoluble compounds $\text{Mg}(\text{OH})_2(s)$ and $\text{CaCO}_3(s)$. Both compounds neutralize the acid present in the gastric juice, as shown in Reactions 1 and 2.



Reaction 1



Reaction 2

The neutralizing capacity of an antacid can be determined by dissolving an antacid tablet in an excess amount of $\text{HCl}(aq)$ of known concentration and titrating the unreacted $\text{HCl}(aq)$ with $\text{NaOH}(aq)$. In an experiment, an antacid tablet was placed in a flask containing 50 mL of 1.5 M $\text{HCl}(aq)$. After the antacid tablet had completely reacted, the solution was gently heated to remove any $\text{CO}_2(g)$. The degassed solution was then titrated using 0.50 M $\text{NaOH}(aq)$ in the presence of an indicator. Figure 1 shows the results obtained from the titration.

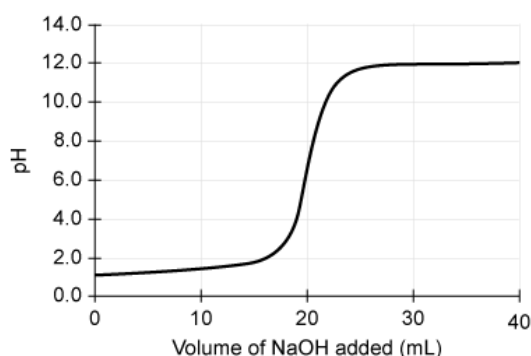


Figure 1 Titration of excess $\text{HCl}(aq)$ with $\text{NaOH}(aq)$

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$$n_{\text{HCl total}} = n_{\text{HCl,antacid}} + n_{\text{HCl,titration}}$$

Equation 1

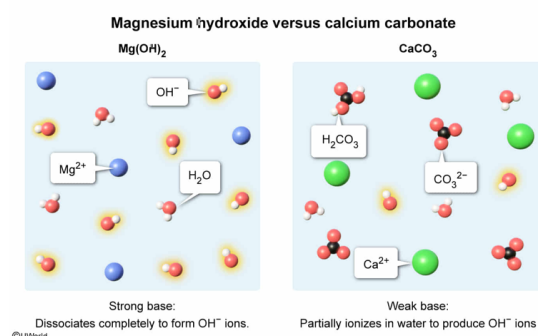
Considering the base components of the antacid, which of the following statements is true?

- ☐ A. $\text{Mg}(\text{OH})_2$ is a weak base because the amount that dissolves in water only partially dissociates into ions. [35%]
- ✓ ☒ B. $\text{Mg}(\text{OH})_2$ is a strong base because the amount that dissolves in water dissociates completely into ions. [57%]
- ☐ C. CaCO_3 is a weak base because the amount that dissolves in water completely ionizes by hydrolysis to form OH^- ions. [3%]
- ☐ D. CaCO_3 is a strong base because the amount that dissolves in water only partially ionizes by hydrolysis to form OH^- ions. [3%]

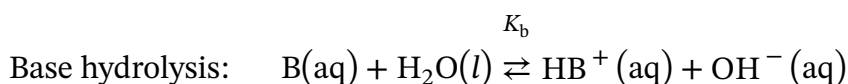
Correct

57% answered correctly

Explanation:



Bases are molecules that can accept one or more acidic protons (H^+ ions). When a base (B) dissolves in water, some of the molecules may remain unchanged while others ionize to form $\text{OH}^-(\text{aq})$ ions in the solution either by direct **dissociation** into ions (eg, an ionic hydroxide base) or indirectly by hydrolysis (ie, taking a proton from water).

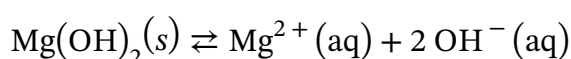


The extent to which the **base molecules ionize** and form $\text{OH}^-(\text{aq})$ ions in water is indicated by the **base dissociation constant K_b** of that base.

$$K_b = \frac{[\text{HB}^+][\text{OH}^-]}{[\text{B}]}$$

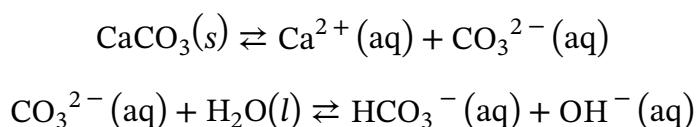
The **larger the K_b** , the greater the tendency to ionize (ie, greater base strength).

Strong bases (typically **Group 1** or **Group 2** metal hydroxides) dissociate completely in water and produce $\text{OH}^-(\text{aq})$. Although sparingly soluble in water, the amount of $\text{Mg}(\text{OH})_2(\text{s})$ that does dissolve dissociates to directly produce $\text{OH}^-(\text{aq})$ ions:



Therefore, $\text{Mg}(\text{OH})_2(\text{s})$ is a **strong base** because the amount that dissolves in water **dissociates completely** into ions.

In contrast, $\text{CaCO}_3(\text{s})$ is a **weak base** that first dissociates into ions and then via hydrolysis, $\text{CO}_3^{2-}(\text{aq})$ produces $\text{OH}^-(\text{aq})$ to a **limited extent**:



(Choice A) $\text{Mg}(\text{OH})_2(\text{s})$ is a strong base and dissociates completely in water.

(Choices C and D) $\text{CaCO}_3(\text{s})$ is a weak base; consequently, it only partially ionizes in water to form $\text{OH}^-(\text{aq})$.

Educational objective:

A base is a substance that can produce hydroxide (OH^-) ions in water either directly through dissociation or indirectly via ionization. The strength of a base depends on the extent to which it produces OH^- in water.

Time spent: 0 sec

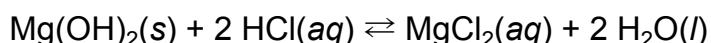
Subject:
General Chemistry

QID: 403934

System:
Solutions and Electrochemistry

Gastroesophageal reflux disease (GERD) is a condition that occurs when acidic gastric juice flows upward from the stomach into the esophagus. This disease is a result of the lower esophageal sphincter's (LES) inability to close. When functioning properly, the LES, which separates the stomach from the esophagus, opens when food is ingested and then closes. If the LES fails to close, gastric juice rushes back into the esophagus, causing tissue irritation and damage.

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Reaction 1



Reaction 2

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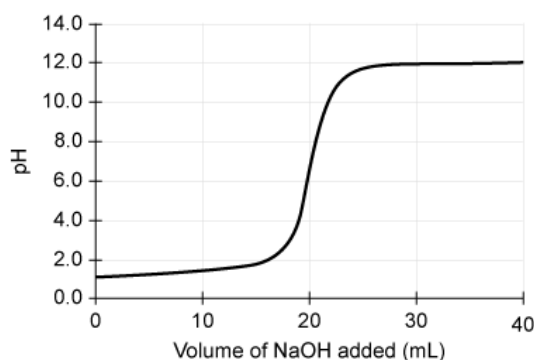


Figure 1 Titration of excess $\text{HCl}(aq)$ with $\text{NaOH}(aq)$

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$$n_{\text{HCl total}} = n_{\text{HCl,antacid}} + n_{\text{HCl,titration}}$$

Equation 1

Indicator	pH range
Thymol blue	1.2–2.8
Methyl red	4.4–6.2
Bromothymol blue	6.2–7.6
Phenolphthalein	8.3–10.5

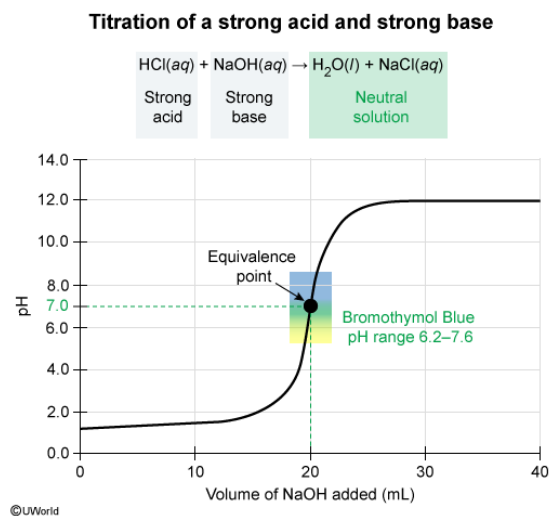
Which of the indicators above is the best to use in the titration in the passage?

- ☐ A. Thymol blue [9%]
☐ B. Methyl red [13%]
☒ C. Bromothymol blue [71%]
☐ D. Phenolphthalein [6%]

Correct

71% answered correctly

Explanation:



During an **acid-base titration** involving a **strong acid** and a **strong base**, a titrant solution with a known concentration of a strong base (or acid) is added to another solution containing an unknown concentration of a strong acid (or base). As the titration proceeds, an **acid-base neutralization** reaction occurs, which changes the **pH** of the solution being titrated. When the solution is fully neutralized, the number of **equivalents** of strong acid is **equal** to the number of equivalents of strong base added (ie, the **equivalence point**), resulting in a **pH of 7**.

Indicators are used to visually detect when an acid-base titration is complete by undergoing a **color change** near the pH of the equivalence point. Different indicators change color across a particular pH range. As such, the best indicator for a titration is one whose pH range most closely overlaps with the pH of the equivalence point.

In the titration of $\text{HCl}(aq)$ (a strong acid) with $\text{NaOH}(aq)$ (a strong base), the equivalence point occurs at a pH of 7. Of the four indicators listed, only bromothymol blue has a pH range that overlaps with the equivalence point and is the best indicator for the titration.

(Choices A and B) Both thymol blue and methyl red change color *before* the equivalence

point has been reached.

(Choice D) Phenolphthalein changes color after the equivalence point has been reached.

Educational objective:

During an acid-base titration, an indicator is used to visually detect the equivalence point of the titration (ie, when the titration is complete). The best indicator for a particular titration is one whose pH range best overlaps with the pH at the equivalence point.

Time spent:0 sec

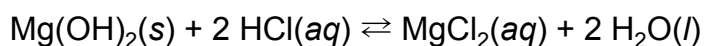
Subject:
General Chemistry

QID:403935

System:
Solutions and Electrochemistry

Gastroesophageal reflux disease (GERD) is a condition that occurs when acidic gastric juice flows upward from the stomach into the esophagus. This disease is a result of the lower esophageal sphincter's (LES) inability to close. When functioning properly, the LES, which separates the stomach from the esophagus, opens when food is ingested and then closes. If the LES fails to close, gastric juice rushes back into the esophagus, causing tissue irritation and damage.

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Reaction 1



Reaction 2

The neutralizing capacity of an antacid can be determined by dissolving an antacid tablet in an excess amount of $\text{HCl}(aq)$ of known concentration and titrating the unreacted $\text{HCl}(aq)$ with $\text{NaOH}(aq)$. In an experiment, an antacid tablet was placed in a flask containing 50 mL of 1.5 M $\text{HCl}(aq)$. After the antacid tablet had completely reacted, the solution was gently heated to remove any $\text{CO}_2(g)$. The degassed solution was then titrated using 0.50 M $\text{NaOH}(aq)$ in the presence of an indicator. Figure 1 shows the results obtained from the titration.

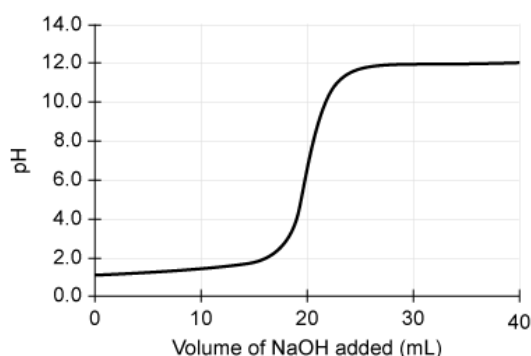


Figure 1 Titration of excess $\text{HCl}(aq)$ with $\text{NaOH}(aq)$

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$$n_{\text{HCl total}} = n_{\text{HCl,antacid}} + n_{\text{HCl,titration}}$$

Equation 1

How many moles of HCl(aq) were neutralized by the antacid tablet?

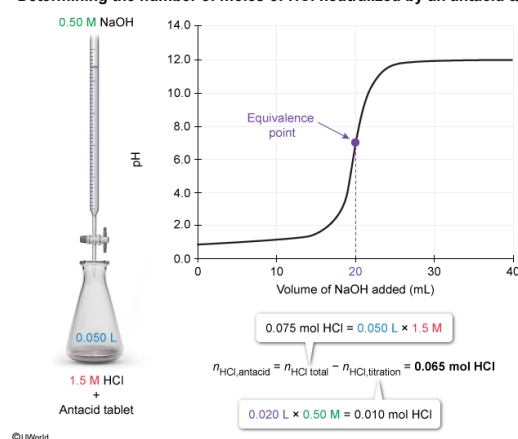
- ☐ A. 0.010 moles [14%]
☒ B. 0.065 moles [39%]
☐ C. 0.075 moles [42%]
☐ D. 0.085 moles [4%]

Correct

38% answered correctly

Explanation:

Determining the number of moles of HCl neutralized by an antacid tablet



During the **titration** of an **acid with a base**, the **equivalence point** occurs when all acidic protons (ie, H^+ ions) have reacted with the base (ie, equivalents of added base = equivalents of initial acid). On a **titration curve**, this is found at the midpoint of the nearly vertical segment.

When the antacid tablet is dissolved in excess HCl(aq) , the antacid **neutralizes** some of the HCl(aq) , and the remaining HCl(aq) is then titrated with NaOH(aq) as described in the passage. The number of moles of HCl(aq) neutralized by the antacid ($n_{\text{HCl,antacid}}$) can be calculated by subtracting the number of moles of HCl(aq) neutralized by NaOH(aq) ($n_{\text{HCl,titration}}$) from the total number of moles of HCl(aq) initially present in the flask ($n_{\text{HCl total}}$):

$$n_{\text{HCl,antacid}} = n_{\text{HCl total}} - n_{\text{HCl,titration}}$$

The total number of moles of HCl(aq) initially present can be calculated by multiplying the **molar concentration** of the HCl(aq) by its volume (converted to liters):

$$n_{\text{HCl total}} = 50 \text{ mL HCl} \times \frac{1 \text{ L}}{1000 \text{ mL}} \times \frac{1.5 \text{ mol HCl}}{\text{L}} = 0.075 \text{ mol HCl}$$

According to Figure 1, the equivalence point is reached after 20 mL of NaOH(aq) has been added to the flask containing the neutralized antacid and unreacted HCl(aq) . Multiplying this volume (converted to liters) by the molar concentration of NaOH(aq) yields the number of moles of NaOH(aq) added:

$$20 \text{ mL NaOH} \times \frac{1 \text{ L}}{1000 \text{ mL}} \times \frac{0.50 \text{ mol NaOH}}{\text{L}} = 0.010 \text{ mol NaOH}$$

Because the neutralization reaction between HCl(aq) and NaOH(aq) occurs in a 1:1 mole ratio, an equal number of moles of HCl(aq) must have been neutralized at the equivalence point:

$$n_{\text{HCl, titration}} = 0.010 \text{ mol NaOH} \times \frac{1 \text{ mol HCl}}{1 \text{ mol NaOH}} = 0.010 \text{ mol HCl}$$

Therefore, 0.065 moles of HCl(aq) were neutralized by the antacid tablet:

$$n_{\text{HCl, antacid}} = 0.075 \text{ mol} - 0.010 \text{ mol} = 0.065 \text{ mol HCl}$$

(Choice A) A quantity of 0.010 moles of HCl(aq) were neutralized by the NaOH(aq) *titrant*, *not* by the antacid.

(Choice C) A *total* of 0.075 moles of HCl(aq) were initially present before adding the antacid to the solution. The number of moles of HCl neutralized by the antacid must be *less* than the total number of moles HCl(aq) initially present.

(Choice D) A value of 0.085 moles results from *adding* the number of moles of HCl(aq) neutralized by NaOH(aq) to the total number of moles of HCl(aq) , rather than *subtracting*.

Educational objective:

In an acid-base titration, the equivalence point occurs when the number of equivalents of acid and base are equal. The equivalence point occurs at the midpoint on the steep slope of the titration curve.

Time spent: 0 sec

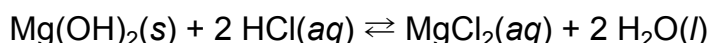
Subject:
General Chemistry

QID: 403936

System:
Solutions and Electrochemistry

Gastroesophageal reflux disease (GERD) is a condition that occurs when acidic gastric juice flows upward from the stomach into the esophagus. This disease is a result of the lower esophageal sphincter's (LES) inability to close. When functioning properly, the LES, which separates the stomach from the esophagus, opens when food is ingested and then closes. If the LES fails to close, gastric juice rushes back into the esophagus, causing tissue irritation and damage.

Antacids are medications that relieve symptoms of GERD by reducing the acidity of the gastric juice, which on average contains 5×10^{-4} moles of $\text{HCl}(aq)$ per 50 mL of gastric juice. A typical antacid contains the water insoluble compounds $\text{Mg}(\text{OH})_2(s)$ and $\text{CaCO}_3(s)$. Both compounds neutralize the acid present in the gastric juice, as shown in Reactions 1 and 2.



Reaction 1



Reaction 2

The neutralizing capacity of an antacid can be determined by dissolving an antacid tablet in an excess amount of $\text{HCl}(aq)$ of known concentration and titrating the unreacted $\text{HCl}(aq)$ with $\text{NaOH}(aq)$. In an experiment, an antacid tablet was placed in a flask containing 50 mL of 1.5 M $\text{HCl}(aq)$. After the antacid tablet had completely reacted, the solution was gently heated to remove any $\text{CO}_2(g)$. The degassed solution was then titrated using 0.50 M $\text{NaOH}(aq)$ in the presence of an indicator. Figure 1 shows the results obtained from the titration.

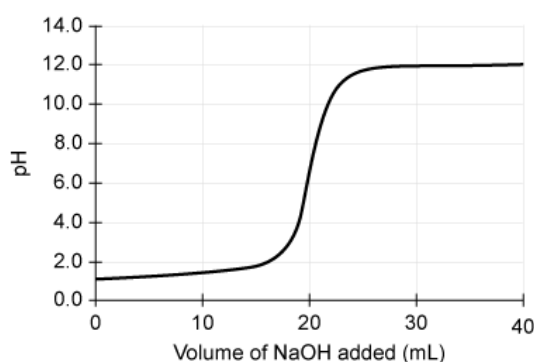


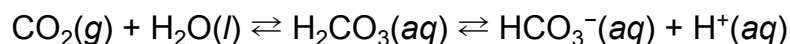
Figure 1 Titration of excess $\text{HCl}(aq)$ with $\text{NaOH}(aq)$

The total number of moles of $\text{HCl}(aq)$ initially present in the beaker, $n_{\text{HCl, total}}$, is equal to the sum of the number of moles of $\text{HCl}(aq)$ neutralized by the antacid tablet, $n_{\text{HCl, antacid}}$ and the number of moles of excess $\text{HCl}(aq)$ neutralized during the titration by $\text{NaOH}(aq)$, $n_{\text{HCl, titration}}$.

$$n_{\text{HCl, total}} = n_{\text{HCl, antacid}} + n_{\text{HCl, titration}}$$

Equation 1

$\text{CO}_2(g)$ dissolves in water and establishes the equilibrium shown below.



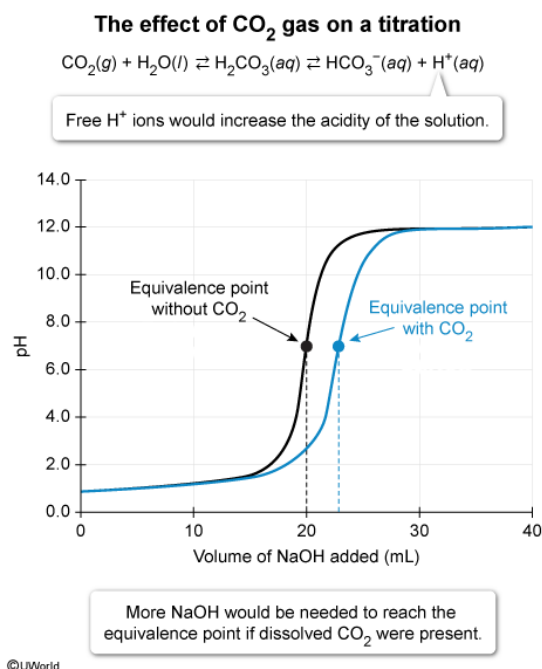
If the researcher neglects to gently heat the antacid solution before titrating, how would the results of the titration be affected?

- ☐ A. Less $\text{NaOH}(aq)$ would be needed to reach the equivalence point because $\text{CO}_2(g)$ would increase the acidity of the solution. [4%]
- ☐ B. Less $\text{NaOH}(aq)$ would be needed to reach the equivalence point because $\text{CO}_2(g)$ would decrease the acidity of the solution. [12%]
- ☐ C. More $\text{NaOH}(aq)$ would be needed to reach the equivalence point because $\text{CO}_2(g)$ would decrease the acidity of the solution. [11%]
- ✔ ☒ D. More $\text{NaOH}(aq)$ would be needed to reach the equivalence point because $\text{CO}_2(g)$ would increase the acidity of the solution. [71%]

Correct

71% answered correctly

Explanation:

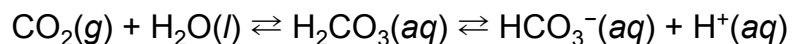


During an **acid-base titration**, a solution with a known concentration of a **base (or acid)** (ie, the titrant) is added to another solution containing an unknown concentration of an acid (or base) (ie, the analyte). At the **equivalence point**, the acid (or base) analyte is fully **neutralized** (ie, the number of **equivalents** of acid equals the number of equivalents of base). On a **titration curve**, the equivalence point is found at the midpoint of the nearly vertical segment.

According to Reaction 2, when $\text{CaCO}_3(aq)$ is neutralized by $\text{HCl}(aq)$, $\text{CO}_2(g)$ is produced:



When $\text{CO}_2(g)$ is dissolved in water, $\text{H}_2\text{CO}_3(aq)$ is formed, and the following equilibrium is established:



The dissociation of $\text{H}_2\text{CO}_3(aq)$ produces free $\text{H}^+(aq)$ ions, which increases the acidity of the solution. Therefore, if $\text{CO}_2(g)$ is not removed from the antacid solution via gentle heating, more $\text{NaOH}(aq)$ would be needed to neutralize the solution (ie, reach the equivalence point) during the titration. This would introduce experimental error in the determination of the amount of $\text{HCl}(aq)$ neutralized by the antacid because the $\text{H}_2\text{CO}_3(aq)$ acid would be mistaken for additional $\text{HCl}(aq)$ in the titration.

(Choices A, B, and C) $\text{CO}_2(g)$ dissolves in water to produce $\text{H}^+(aq)$ ions, which *increases* the acidity of the antacid solution. Consequently, *more* $\text{NaOH}(aq)$ would be needed to neutralize the solution.

Educational objective:

Carbon dioxide forms carbonic acid when dissolved in water, which acidifies the solution. The increased acidity from the carbonic acid alters the results of a titration.

Time spent:0 sec

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